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Palaeomagnetism and K–Ar age of the Upper Ordovician Alcaparrosa Formation, Argentina

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Summary. Palaeomagnetic data from 71 hand samples of igneous rocks of Late Ordovician age exposed in western Argentina (31.3°S, 69.4°W, Alcaparrosa Formation) are given. Stable remanent magnetization was isolated in the majority of samples; they yield a palaeomagnetic pole at 56°S 33°E ($N = 8$, $\alpha_{95} = 16°$). Whole rock K–Ar age determinations yield an age of 416 ± 10 Myr for a pillow lava of the Alcaparrosa Formation.

Palaeomagnetic data for South America, Africa, Australia, Antarctica and India suggest that Gondwana was a unit at least as far back as 1000 Myr. The palaeomagnetic data define a rapid polar migration for Gondwana in Ordovician time which is consistent with the widespread occurrences of Late Ordovician glacial deposits across this supercontinent.

Introduction

The distribution of continents and oceans in Late Precambrian and Early Palaeozoic is one of the key problems in Earth sciences and is now the subject of debate. Palaeomagnetic data when combined with geological criteria, with faunal distributions and with palaeoclimatic indicators may indicate the generation and destruction of former oceans. However, for the Late Precambrian and Early Palaeozoic times the accuracy and the distribution of control points on apparent polar wandering curves prevent unequivocal solutions (McElhinny & Embleton 1976). To provide data points for South America the palaeomagnetic laboratories of the Universities of Buenos Aires (Argentina) and São Paulo (Brazil) have programmed the study of sequences of South American rocks assigned to the Late Precambrian and Early Palaeozoic. In this paper we present the palaeomagnetic data and the isotopic age from a sequence of igneous rocks (Alcaparrosa Formation) in western Argentina assigned to the Late Ordovician. We also compare the reliable palaeomagnetic poles for South America assigned to the Late Ordovician (this paper), Cambrian (Thompson 1973) and Late Precambrian (Vilas & Pacca 1978), with results from the rest of Gondwana.

Geological evidence and sampling

Marine sediments and basic rocks included in the Alcaparrosa Formation are exposed in the western part of the Province of San Juan, Argentina (31.3°S 69.4°W, Fig. 1) (Quartino,

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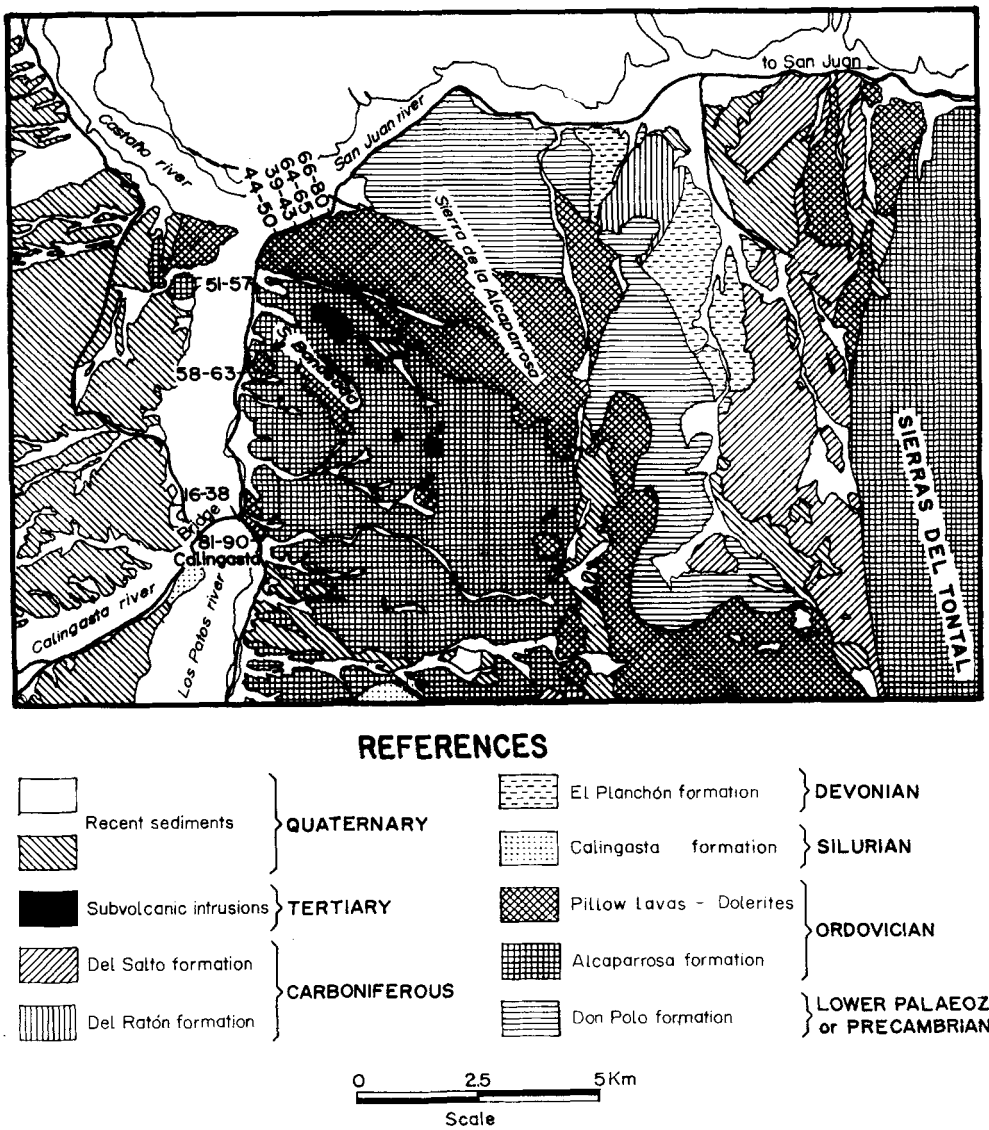


Figure 1. Schematic geological map near Calingasta (Quartino *et al.* 1971) showing sampling sites for the palaeomagnetic and radiometric studies.

Zardini & Amos 1971). The sediments of the Alcaparrosa Formation are characterized by the presence of fossil assemblages (graptolites) from which Turner (1960) has suggested a Caradocian age for this formation. The basic rocks include dolerites and basalts with pillow lavas formed during a tectomagmatic eugeosynclinal phase related to Ordovician sedimentation (Quartino *et al.* 1971). The pillow lavas are well exposed along the route between Calingasta and San Juan. In Sierra Bandeada (sites 9058/9063 in Fig. 1) nine lava flows are exposed. In this site and in the sites situated further north the dip of the pillow lavas is of about 30°; the pillow lavas exposed close to the bridge of Calingasta (site 9016/9038, Fig. 1) are subvertical (80°).

The igneous rocks of the Alcaparrosa Formation were sampled at eight different sites; 75 hand samples were collected (Fig. 1). Whenever possible, samples were collected at

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Table 1. K–Ar ages for the two oldest pillow lavas of the Alcaparrosa Formation exposed close to the bridge of Calingasta.

Sample number	K (per cent)	Ar^{40}_{rad} 10^{-10} mol/g	Ar^{40}_{atm}	Age (Myr)
9026	0.21	1.741	33.1	416 ± 10
9017	0.23	1.307	46.3	294 ± 15

different stratigraphic levels at each site. Hand samples were oriented using both Sun and magnetic compasses. Observations of the bedding plane were carried out in each of the sampling sites. Four samples were destroyed during transport.

Potassium–argon determinations

Whole potassium–argon age determinations were carried out on two hand samples collected from two basalts (pillow lavas) exposed close to the bridge of Calingasta (site 9016/9038, Fig. 1). Four lava flows are exposed here. The study was carried out on samples numbers 9017 and 9026 collected from the two lowest pillow lavas exposed in the site, in the Instituto de Geocronología y Geología Isotópica (Universidad de Buenos Aires). The techniques used are those indicated by Linares & Valencio (1975). The K–Ar data are shown in Table 1. In view of the palaeontologic data the age obtained from the sample 9017 (the oldest lava flow) is considered to be anomalously affected by argon loss. Sample 9026 yields an age of 416 ± 19 Myr indicative of Late Ordovician–Early Silurian age.

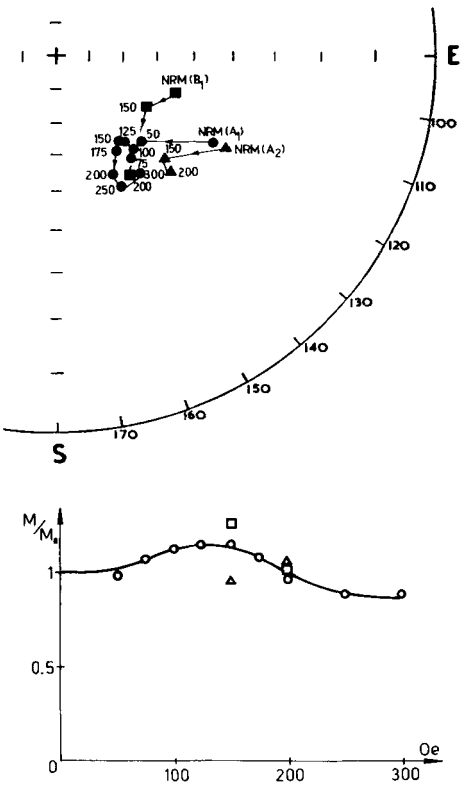


Figure 2. Changes in direction (referred to the palaeohorizontal) and intensity of NRM of the pilot specimen and two other cylinders from one hand sample (9046) from Alcaparrosa Formation in response to AF treatment.

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Table 2. Summary of palaeomagnetic results of the Alcaparrosa rocks exposed in western Argentina.

Geological unit or site	Sample numbers	Cleaned remanent magnetization							South virtual geomagnetic pole			
		Referred to present horizontal				Referred to palaeohorizontal						
		N	D (°)	I (°)	α_{95} (°)	k	D (°)	I (°)	Latitude (°)	Longitude E (°)	$d\psi$ (°)	$d\chi$ (°)
Los Patos, A	9081/84	3	1.1	−46.0	47.9	7.6	354.5	−5.8	–	–	–	–
Los Patos, B	9085/90	4	147.1	−9.5	89.0	2.0	139.9	−49.9	–	–	–	–
Km 126 and 125.5	9064/80	16	152.0	41.3	14.4	7.5	113.0	62.2	36.9 S	12.7	17.4	22.4
Km 126.5	9039/43	5	143.3	5.9	22.7	12.3	134.6	28.9	45.7 S	30.9	13.8	25.0
Km 127.1	9044/50	7	145.5	43.3	12.5	23.9	103.3	59.8	29.8 S	348.7	14.2	18.8
Quebrada de la Casita	9051/57	7	48.6	−38.1	19.0	10.9	25.1	−65.4	64.9 N	67.9	25.0	30.8
Sierra Bandeada	9058/63	4	132.8	48.9	21.3	19.5	144.6	33.7	55.5 S	34.1	13.8	24.3
4° lava flow (younger)	9035/38	3	103.2	23.5	17.6	49.6	158.7	34.7	67.2 S	48.0	11.6	20.2
3° lava flow	9029/34	6	143.9	33.9	12.2	30.8	171.5	0.0	57.6 S	94.5	6.1	12.2
2° lava flow	9022/28	5	116.7	22.9	28.1	8.3	157.7	22.4	61.4 S	59.4	15.8	29.8
1° lava flow (older)	9016/21	6	106.0	20.6	34.7	4.6	155.1	32.3	63.5 S	46.3	22.1	39.1

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The palaeomagnetic study

Two hundred and twenty-five cylinders of 2.5-cm diameter and height were cut. Two spinner magnetometers (one a Digico and one made in the Universidad de Buenos Aires) were used to measure the remanent magnetization. Detailed alternating-field demagnetization was used to isolate the stable remanence of one pilot cylinder of each sample (Valencio, Vilas & Mendía 1977). Two other cylinders from the same sample were submitted to the optimum alternating field so as to improve the definition of the direction of the stable remanence of each hand sample.

A secondary component of remanence was present in all the samples; it has the characteristics of a viscous remanent magnetization and was destroyed by the alternating demagnetization. The optimum demagnetizing AC field varied between 100 and 150 Oersteds at which more than 70 per cent of the NRM intensity was retained. Fig. 2 shows the changes in direction and intensity of NRM of one pilot specimen from Alcaparrosa Formation in response to AF treatment. It was not possible to isolate stable remanence in five samples; one of these is number 9017, which yielded an anomalous isotopic age (Table 1).

The method of Fisher (1953) was used to compute mean directions on natural (NRM) and cleaned remanent magnetization of either each lava flow or sampling site assigning unit weight to the mean direction of the natural and cleaned remanences of each of its hand samples, respectively. Lava flow and site-mean directions of cleaned remanence were referred to the present and palaeohorizontal (Table 2). The scatter of directions of cleaned remanence of samples from Los Patos A and B are larger than 40°; therefore we do not include these directions in the computed mean direction for the Alcaparrosa Formation (Table 3). This table shows that the site-mean directions and the lava flow-directions are better grouped after bedding-plane correction suggesting that the cleaned remanence of these units (Fig. 3) was acquired before the folding of the beds.

Virtual geomagnetic poles for each site and lava flow were calculated (Table 2). A palaeomagnetic pole was calculated by meaning these virtual geomagnetic poles; we rejected the VGP's situated more than 40° from the palaeomagnetic pole and a new mean pole was computed. In this way a population of VGP's was obtained all within 40° of the palaeomagnetic pole; this new population of VGP's gives the palaeomagnetic pole for the Alcaparrosa Formation at 56.2° S 32.8° E ($N = 8$, $k = 12.2$, $\alpha_{95} = 16.4°$, $SAOu_1$). Fig. 4 shows the position of $SAOu_1$. The rejected VGP (Quebrada de las Casitas) is classified as oblique.

Interpretation of results

When considering palaeomagnetic results in plotting apparent polar wander paths and making palaeogeographic reconstructions it is necessary to consider the effects of the polar position errors and the accuracy of the age assigned to the rocks. The technique used here is similar to that used by McElhinny & Embleton (1976). However, we use a more stringent criterion

Table 3. Mean directions of cleaned remanence of the Alcaparrosa rocks.

Mean cleaned remanent magnetization									
Referred to present horizontal					Referred to palaeohorizontal				
N	D (°)	I (°)	α_{95} (°)	k	N	D (°)	I (°)	α_{95} (°)	k
9	136.4	35.5	21.8	6.4	9	150.6	40.2	19.0	8.2

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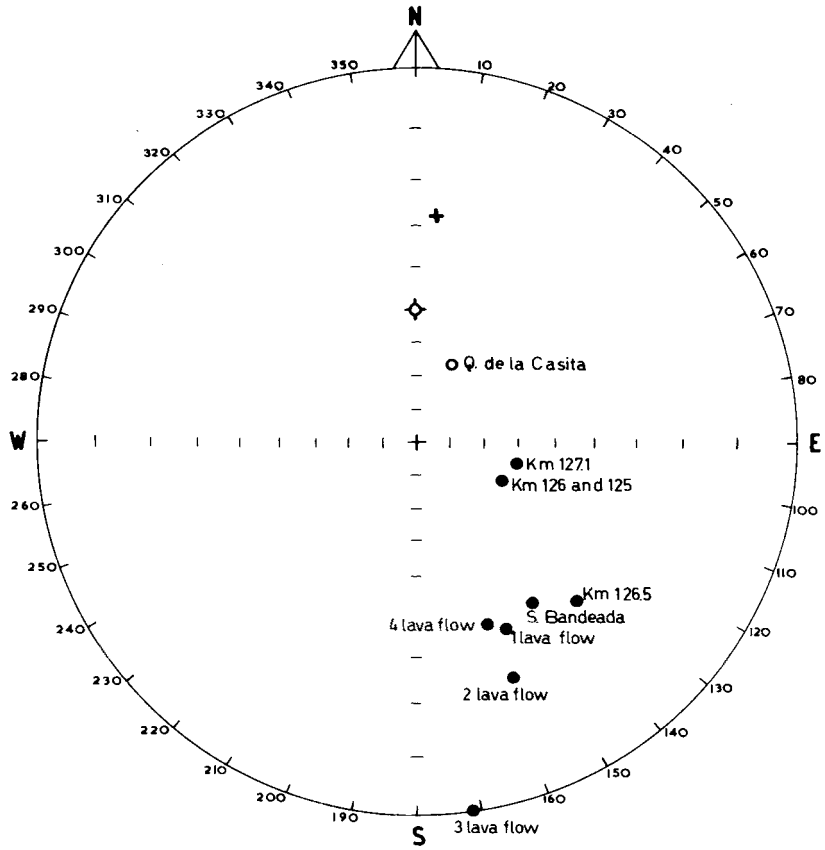


Figure 3. Alcaparrosa Formation. Mean directions of cleaned remanent magnetization of lava flows and sites referred to the palaeohorizontal. Solid symbols indicate downward dipping directions. The directions of the present geomagnetic field and the axial dipole are shown by + and ⟡ respectively.

for the palaeomagnetic data to be sure that the study includes samples collected covering a minimum time span to average out the non-dipole components of the geomagnetic field (Table 4). Following those authors the points allocated to each palaeomagnetic pole position are added to the points allocated to the age of the rocks to give a maximum of six points. According to the total points allocated to each determination pole positions are given the following classifications: six points, key poles; five points, main poles; three or four points, subsidiary poles and two or three points, supporting poles.

Pole positions and age information for Late Precambrian (<1000 Myr) and Early Palaeozoic rocks for South America are listed in Table 5 together with their classifications according to the scheme adopted. Fig. 4 shows the position of the key, main and subsidiary poles for South America; it also shows the Late Palaeozoic–Early Mesozoic section of the apparent polar wandering curve for this continent from the Moscovian (SAC_2) to the late Early Cretaceous–Late Cretaceous quasi-static period (SAK1-u) (Vilas & Valencio 1978 and Valencio *et al.* 1977). The figure shows that the palaeomagnetic poles are not sufficient to define the apparent polar path for South America in Late Precambrian and Early Palaeozoic times. This is more evident when they are compared with the joint apparent polar wandering path for the Gondwana (see below). Therefore we can not evaluate the position of the Alcaparrosa pole ($SAOu_1$) on the basis of the available palaeomagnetic data for South America alone.

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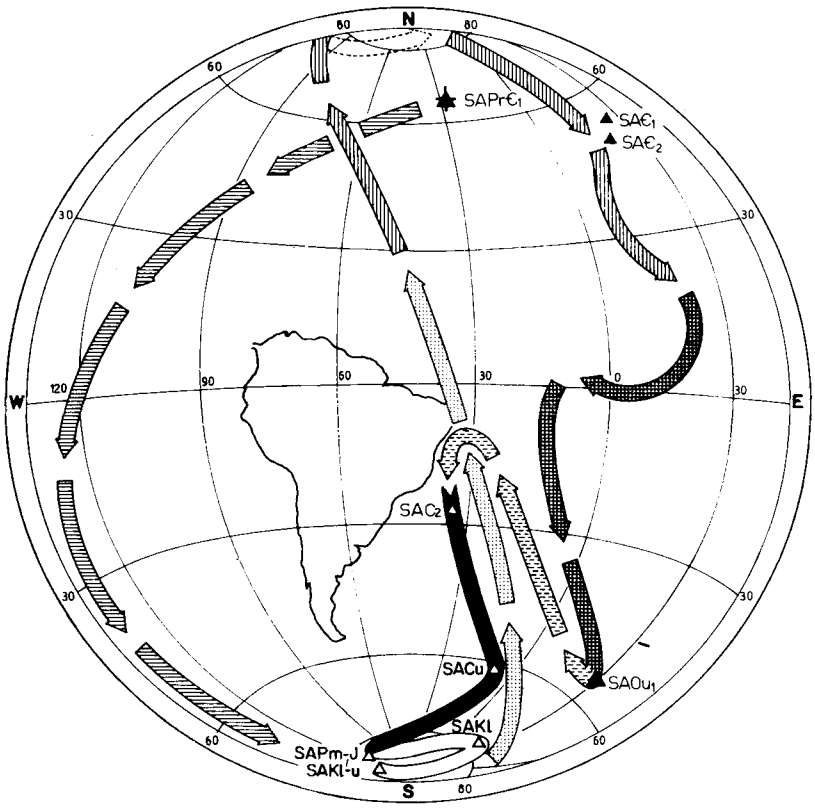


Figure 4. Pole positions for South America from 1000 to 400 Myr (Table 5): $SAOu_1$ is the pole for the Alcaparrosa Formation. The symbols are plotted in three forms depending on the total classification (see in the text): ★, six points; △, five points; ▲, four or three points. The arrows show the joint apparent polar path for the Gondwana (Fig. 5); the age of the polar path is as follows: ▤Precambrian from 1000 to 700 Myr; ⋯ Late Precambrian; ||||| Cambrian; ### Ordovician; ⋰⋱ from Late Ordovician to Late Carboniferous. Other references in the text. (Lambert's equal-area projection.)

Table 4. Classification system for palaeomagnetic data.

Palaeomagnetic classification		Points (*P*)
Stratigraphic sequence of samples not known	Sediments: 10 samples; 1 site; stability test; $\alpha_{95} < 25°$ or $k > 15$	1
	Igneous rocks: 10 samples; 2 sites or 3 time units; stability test; $\alpha_{95} < 25°$ or $k > 15$	
Stratigraphic sequence of samples known	Sediments: >10 samples; more than 10 m of sequence; stability test; $\alpha_{95} < 25°$ or $k > 15$ Igneous rocks: >10 samples; 4 sites or 5 time units, stability test; $\alpha_{95} < 25°$ or $k > 15$	2
	Sediments: >10 samples; more than 10 m of sequence; stability test; $\alpha_{95} < 15°$ Igneous rocks: >10 samples; 4 sites or 5 time units; stability test; $\alpha_{95} < 15°$	3

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Table 5. Palaeomagnetic pole positions from South America, Africa, Antarctica, Australia and India for the period 1000–400 Myr. (The rotated pole positions are those for each continent rotated to the Vilas & Valencio (1977) reconstruction of Gondwana and with Africa remaining fixed in its present-day coordinates.)

Rock unit or sampling area	Classification P + A	Age	Symbol	Pole position (°)	(°)	Rotated pole position (°)	(°)	Reference
South America								
Urucum Formation	1 + 1	O–S	–	34 N	344 E	46 N	1 E	Creer (1965)
Alcaparrosa Formation	2 + 3	Ou	$SAOu_1$	56 S	33 E	24 S	79 E	This paper
Campanario Formation (South Tilcara)	1 + 2	ϵ	$SA\epsilon_2$	52 N	27 E	81 N	27 E	Thompson (1973)
Campanario Formation (North Tilcara)	1 + 2	ϵ	$SA\epsilon_1$	49 N	23 E	77 N	28 E	Thompson (1973)
Bambui Group	3 + 3	886 ± 11 Myr	$SAPr\epsilon_1$	65 N	336 E	60 N	312 E	**Vilas & Pacca (1978)**
Africa								
Sediments and lavas, Moroccan Anti-Atlas	3 + 3	530 Myr	$Afem_3$	47 N	42 E	–	–	Daly & Pozzi (1977)
Doornpoort Formation	2 + 2	550–500 Myr	$Afem_2$	42 N	61 E	–	–	Piper (1975a)
Amouslek Tuffs	2 + 3	ϵ_1	$Afel_1$	41 N	250 E	–	–	Hailwood & Tarling (1973)
Ouarzazate Volcanics	2 + 1	Prϵu	$AfPr\epsilon_9$	30 N	237 E	–	–	Hailwood & Tarling (1973)
Klipheuvel Formation	2 + 1	Pr/ϵ	$AfPr/\epsilon_1$	16 N	316 E	–	–	Creer (1973)
Ntonya Ring Structure	3 + 3	630 ± 24 Myr	$AfPr\epsilon_8$	28 N	345 E	–	–	Briden (1968)
Pre-Nama dykes	2 + 2	653 ± 70	$AfPr\epsilon_7$	85 N	228 E	–	–	Piper (1975a)
Mbozi Complex	1 + 3	743 ± 30	$AfPr\epsilon_6$	72 N	68 E	–	–	Piper (1975b)
Bukoban intrusives	2 + 3	806 ± 30	$AfPr\epsilon_5$	11 N	101 E	–	–	Piper (1972)
Gagwe lavas	3 + 3	813 ± 30	$AfPr\epsilon_4$	29 N	103 E	–	–	Piper (1972)
Bukoban sandstones	1 + 2	>813 <1000	$AfPr\epsilon_3$	40 N	317 E	–	–	Piper (1972)
Malagarasi Sandstone	1 + 2	1000–800	$AfPr\epsilon_2$	7 N	112 E	–	–	**Piper (1972)**
Umkondo lavas	3 + 3	1115	$AfPr\epsilon_1$	62 N	15 E			McElhinny (1966), Piper (1976)

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Arabia								
Jordanian Red beds	1 + 1	ϵ [− 0]	–	37 N	323 E	–	–	Burek (1969)
Antarctica								
Sør Rondane intrusives	1 + 3	01-m 485 ± 25 Myr	AnOl-m_1	28 N	190 E	20 N	20 E	Zijderveld (1968)
Charnockites, Mienyy	3 + 3	ϵu-Ol	Anϵ/O_1	2 N	208 E	51 N	24 E	McElhinny (1973)
Australia								
Mereenie sandstone	2 + 1	S?-D	AuD_1	41 S	40 E	17 N	29 E	Embleton (1972b)
Tumblagooda sandstone	3 + 2	O	AuO_3	30 S	31 E	29 N	23 E	Giddings (1975)
Stairway sandstone	2 + 3	Om	$AuOm_2$	2 S	50 E	50 N	58 E	Embleton (1972b)
Jinduckin Formation	3 + 3	Ol	$AuOl_1$	13 S	25 E	47 N	18 E	Luck (1970, 1972)
Lower Lake Frome Group	3 + 3	ϵm	Auϵm_4	8 S	25 E	52 N	18 E	Embleton & Giddings (1974)
Hudson Formation	1 + 3	ϵm	Auϵm_3	18 N	19 E	77 N	353 E	Luck (1972)
Aroona Dam sediments	2 + 3	ϵ_1	Auϵl_2	36 S	33 E	23 N	24 E	Embleton & Giddings (1974)
Antrim Plateau Volcanics	2 + 3	ϵ_1	Auϵl_1	9 S	340 E	32 N	322 E	McElhinny & Luck (1970)
Arumbera sandstone	2 + 1	Pr/ϵl	AuPr/ϵl_1	8 N	325 E	30 N	296 E	Embleton (1972a)
Pound quartzite	2 + 2	Latest Prϵ	$AuPr\epsilon_2$	60 S	6 E	2 S	8 E	Embleton & Giddings (1974)
YB dykes	1 + 3	750 ± 30 Myr	$AuPr\epsilon_1$	20 S	282 E	20 S	299 E	Giddings (1975)
India								
Salt Pseudomorph Beds	3 + 3	ϵm	Inϵm_2	30 N	354 E	60 N	359 E	Wensink (1972)
Purple sandstone, Salt Range	2 + 3	ϵ_1	Inϵl_1	28 N	346 E	63 N	346 E	McElhinny (1970)
Upper Bhander sandstone	3 + 1	745 Myr/ϵ	$InPr\epsilon_2$	49 N	213 E	24 S	335 E	Klootwijk (1973)
Malani rhyolites	3 + 3	745 ± 10 Myr	$InPr\epsilon_1$	78 N	45 E	47 S	274 E	Athavale, Radhakrishnamurty & Sahasrabudhe (1963)
Kaimur Series	1 + 1	900–1200 Myr		82 N	286 E	53 S	301 E	Sahasrabudhe & Mishra (1966), Crawford & Compston (1970)

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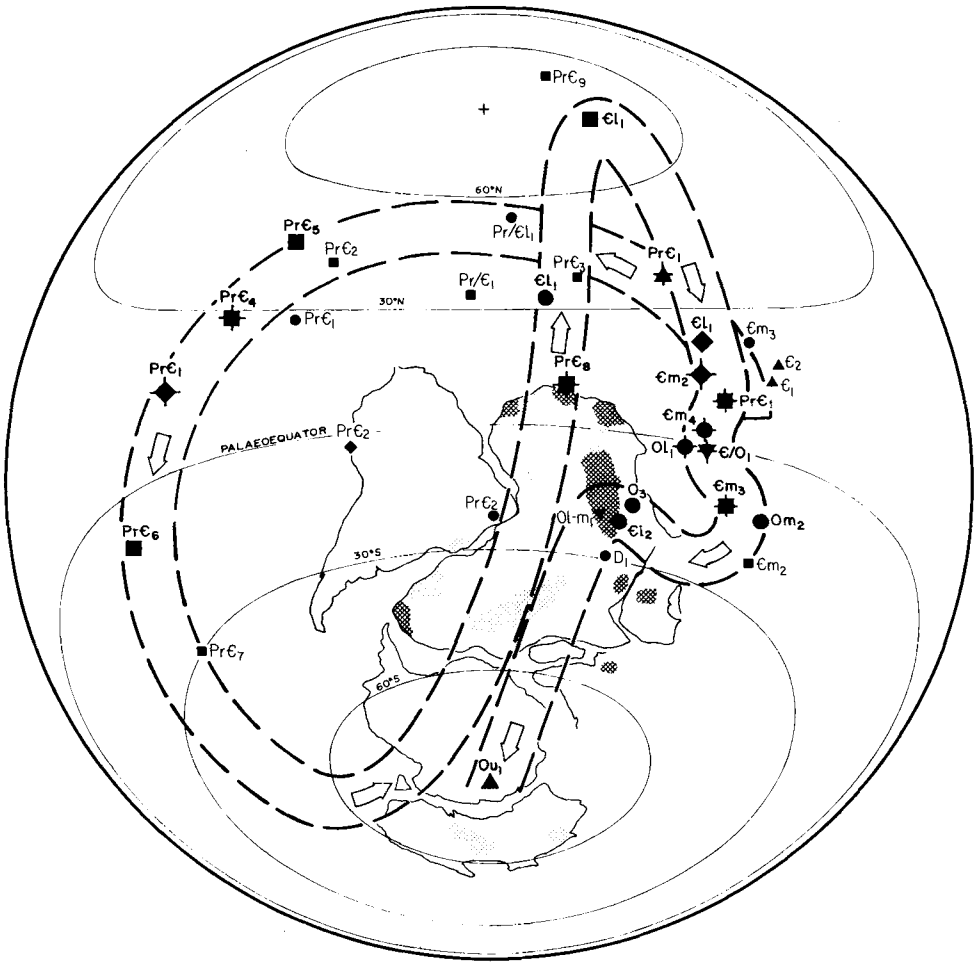


Figure 5. Pole positions for the Gondwanic continents for the period 1000–400 Myr ago plotted on the Vilas & Valencio (1978) reconstruction and joint apparent polar wandering path for the Gondwana. Latitude–longitude grid based on the magnetism of the Alcaparrosa rocks (Late Ordovician). The key of the symbols as in Fig. 4; ▲, South America; ■, Africa; ●, Australia; ◆, India; ▼, Antarctica ▒, Late Precambrian glaciations; ▓, Late Ordovician glaciations, after McElhinny & Embleton (1976). (Lambert's equal-area projection.)

McElhinny & Embleton (1976) showed that the palaeomagnetic data suggest Gondwana (Smith and Hallam reconstruction) was a unit at least as far back as 750 Myr ago. The comparison of the palaeomagnetic pole positions for the Gondwanic continents classified according to the scheme adopted (Table 5) will allow us to evaluate whether the position of the Alcaparrosa pole is consistent with the Ordovician poles of the other Gondwanic continents after reconstruction. To do that we have rotated all the palaeomagnetic poles for the Gondwanic continents for the Late Precambrian–Early Palaeozoic interval to conform with the Vilas & Valencio (1978) reconstruction, Fig. 5. This figure shows that the position of the Alcaparrosa pole does not agree with the other Gondwanic Ordovician poles.

To proceed further, an examination of Fig. 5 shows that the palaeomagnetic poles abridged in Table 5 extend the interpretation that Gondwana was a unit at least as far back

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as 1000 Myr ago. This is based on the good agreement between the well-dated palaeomagnetic poles for the Bambui Group ($SAPr\epsilon_1$, 886 ± 11 Myr) and the Umkondo Lavas ($AfPr\epsilon_1$, 1115 Myr) for South America and Africa, respectively, and the poles $AuPr\epsilon_1$ (750 ± 30 Myr) for Australia, $AfPr\epsilon_4$ (813 ± 30 Myr), $AfPr\epsilon_5$ (806 ± 30 Myr) and $AfPr\epsilon_2$ (1000–800 Myr) for Africa and $InPr\epsilon_1$ (745 ± 10 Myr) for India. Particularly, note the excellent agreement between Cambrian poles from Australia ($Au\epsilon m_4$, $Au\epsilon m_3$), India ($In\epsilon l_1$, $In\epsilon m_2$), South America ($SA\epsilon_1$, $SA\epsilon_2$), Antarctica ($An\epsilon/O_1$) and Africa ($Af\epsilon m_3$). However, we must realize that these palaeomagnetic data do not exclude models involving the opening and closing of minor (intracratonic) oceans.

Fig. 5 shows the joint apparent polar wandering path for Gondwana; we have extended the Late Precambrian section of this path to the southeastern tip of Australia to take account of the palaeomagnetic data for the Marinoan Glacial beds quoted by McElhinny & Embleton (1976). The joint pole path is consistent with the widespread occurrences of Late Precambrian and Late Ordovician glacial deposits across Africa suggesting their timing and location are related to rapid polar migration. Particularly, the polar path defined by the Alcaparrosa pole is broadly consistent with the Late Ordovician glacial deposits in India and in southern Africa.

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